

The Beginnings of Logic

Validity, Demonstration, Dialectic, Sophistry, and Plato

History of Logic

Primary source: William and Martha Kneale,
The Development of Logic, Chapter I, “The Beginnings,”

1 The Historical Problem: When Does Logic Begin?

Human beings certainly made inferences and criticized the inferences of others long before Aristotle. But the mere practice of reasoning is not yet logic in the sense relevant to the history of the subject.

A person can perform an activity correctly without possessing an explicit theory of its rules. In the same way, a person can reason correctly without formulating principles of valid reasoning.

For the Kneales, logic begins when reasoning itself becomes an object of reflection: when thinkers begin asking what makes one inference valid and another invalid.

THE BEGINNINGS OF LOGIC

Reasoning $\neq$ Logic

Logic = reflection on the principles of valid inference

The distinction matters historically. We are therefore not asking:

When did people first reason?

We are asking:

When did people begin to formulate and investigate principles governing correct inference?

Historical question

When did reasoning become an object of study?

2 Proof: Truth and Validity

Logical reflection arises most naturally in contexts where proof is sought or demanded.

To prove a proposition is not merely to produce statements in its favor. Proof requires both appropriate starting points and an appropriate relation between those starting points and the conclusion.

The Kneales distinguish two requirements:

1. the premisses must be true;
2. the inference from those premisses to the conclusion must be valid.

These requirements are logically distinct. An argument may use false premisses while still exhibiting a valid form of inference. Conversely, true premisses do not rescue an invalid inference.

PROOF requires

1. True premisses
2. Valid inference

Truth of premisses $\neq$ validity of inference

This distinction will remain central throughout the history of logic.

3 Three Sources of Early Logical Reflection

The Kneales distinguish three broad kinds of discourse in which proof is characteristically sought.

First, in **mathematics**, thinkers seek proofs of abstract truths.

Second, in **metaphysics**, thinkers seek proofs of very general claims about reality.

Third, in **everyday argument**, especially political and forensic argument, speakers seek to establish contingent claims and defeat opponents.

These three settings produce three major routes into logical reflection.

WHERE DOES LOGICAL REFLECTION ARISE?

Mathematics	→	demonstration
Metaphysics	→	dialectic / refutation
Everyday debate	→	fallacy / sophism

Demand for proof → reflection on inference

Of these three, mathematics corresponds most clearly to what Aristotle later calls demonstrative reasoning. It is therefore natural to begin there.

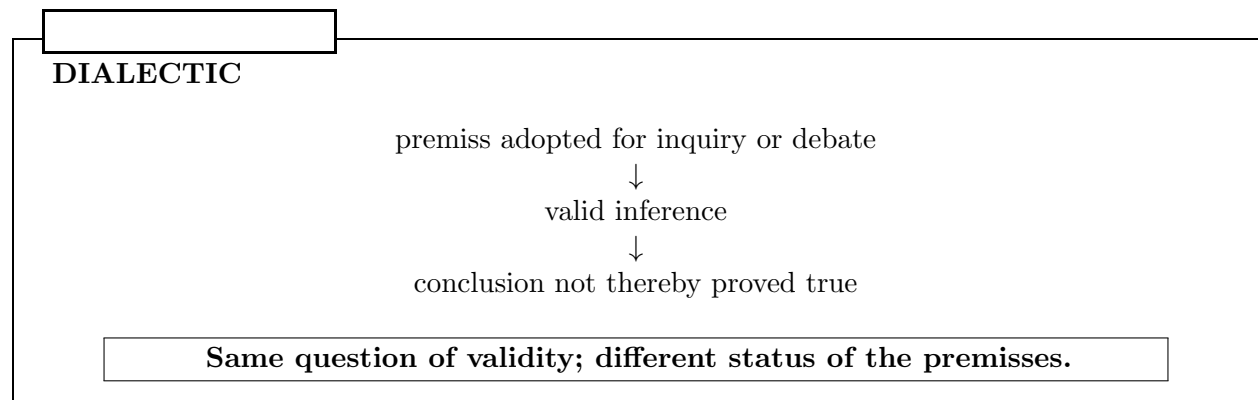
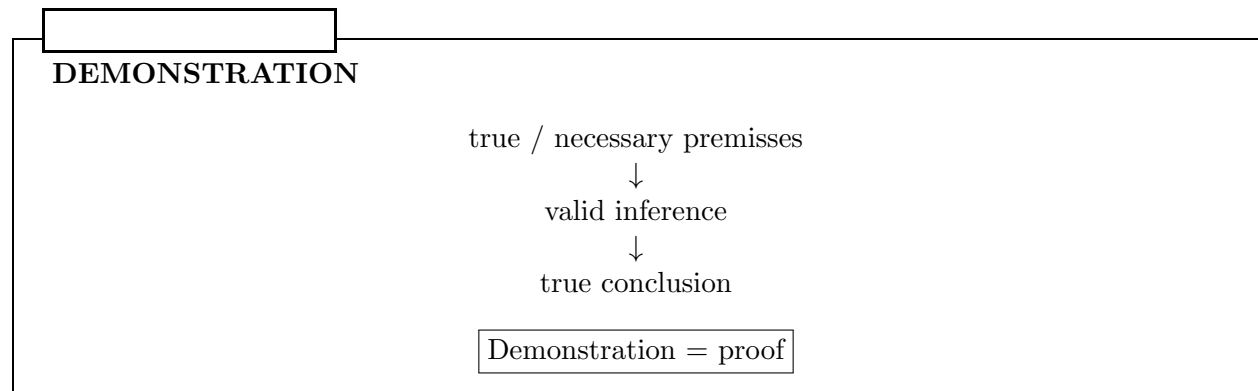
4 Demonstrative and Dialectical Reasoning

Aristotle later distinguishes demonstrative reasoning from dialectical reasoning, and the Kneales use this distinction to help explain the earlier development of logic.

A **demonstrative premiss** is treated as true and necessary. In demonstration, true starting points are connected to a conclusion by necessary inference. This produces proof.

A **dialectical premiss**, by contrast, is adopted in inquiry or debate. It may be plausible or generally accepted without already being established as true.

Thus a dialectical argument may be perfectly serious and may use valid inference, but its conclusion is not thereby demonstrated to be true.



The important point is that the distinction between demonstration and dialectic does not amount to a distinction between valid and invalid reasoning.

5 Geometry and the Rise of Demonstration

The Kneales suggest that the notion of demonstration first attracted sustained attention in connection with geometry.

Egyptian geometry had already produced significant results, including results useful for practical measurement. The decisive Greek development was not simply the discovery of more geometrical facts, but the transformation of geometry into a demonstrative science.

The word “geometry” itself originally reflected its association with land measurement. Greek mathematicians increasingly moved from such empirical procedures toward proofs intended to show that a result *must* be true.

Tradition credits Thales with some of the earliest geometrical proofs, while systematic geometrical study became especially associated with the Pythagorean school.

GEOMETRY

Egyptian geometry $\longrightarrow$ practical / empirical

Greek geometry $\longrightarrow$ demonstrative / deductive

“It is true” $\longrightarrow$ “Why must it be true?”

The importance of geometry for logic lies precisely in this demand for necessity.

6 The Structure of a Deductive Science

The customary presentation of geometry as a deductive science contains three requirements.

First, some propositions must be accepted as starting points without demonstration.

Second, the other propositions of the science must be derived from those starting points.

Third, the derivations must not depend upon additional geometrical assumptions that have never been stated.

The third condition is especially important for logic. Once we require every theorem to follow from explicitly stated starting points, attention is directed toward the relation of **logical consequence** or **entailment**.

DEDUCTIVE SYSTEM

1. Primitive propositions
2. Derived propositions
3. No hidden assumptions

axioms / starting points

$\downarrow$

logical consequence

$\downarrow$

theorem

Key logical question: What follows from what?

Geometry therefore became the historical paradigm of deductive system-building.

7 Seeing a Theorem and Demonstrating It

The early Greek geometers probably did not immediately possess the later ideal of rigorous deduction.

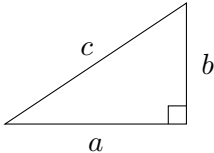
A procedure could count as satisfactory if it enabled one to “see” why a theorem was true. The theorem traditionally associated with Pythagoras provides a good illustration.

For a right triangle with legs a and b and hypotenuse c ,

$$c^2 = a^2 + b^2.$$

Early demonstrations may have used arrangements of congruent triangles from which the equality of the relevant areas appeared visually evident.

PYTHAGOREAN THEOREM



$$c^2 = a^2 + b^2$$

Visual rearrangement can make the result appear evident.

“Looks evident” $\neq$ deductive proof

The issue is not that the visual insight is worthless. The issue is that a rigorous demonstration must establish every geometrical claim on which the reasoning depends.

A diagram may invite us to see that a figure is a square, for example, while a deductive proof must establish that fact from previously accepted principles.

FROM VISUAL INSIGHT TO DEMONSTRATION

stated assumptions



lemma



lemma



theorem

8 The Euclidean Ideal

By the time of Euclid, the ideal of geometrical demonstration had become much clearer.

The aim was to place the specifically geometrical assumptions at the beginning and then construct chains in which later propositions followed from those assumptions by necessity.

The Kneales note that Euclid himself does not always satisfy this ideal perfectly. In the first proposition of the *Elements*, for example, he relies upon an assumption concerning the intersection of two circles without first stating a postulate from which that result follows.

The significance of the example is methodological: if such an unstated assumption were detected, the natural repair would be to make the assumption explicit.

EUCLIDEAN IDEAL

All special assumptions stated explicitly.

hidden assumption



gap in demonstration



state an additional postulate

A deductive system must expose its starting points.

The history of geometry thus creates pressure toward explicit axioms, explicit derivations, and explicit relations of consequence.

9 What Geometry Encouraged Logicians to Notice

If Greek logical reflection developed partly from geometry, the Kneales argue that several later features of Greek logic become understandable.

9.1 General Propositions

Geometry is not fundamentally concerned with one unique individual line or triangle.

When a geometer speaks of “the line AB ,” the object drawn functions as a representative of any line satisfying the relevant conditions.

1. GENERAL PROPOSITIONS

“the line AB ”

really functions as

any line satisfying the stated conditions.

9.2 Necessary Universal Propositions

Geometry also encourages a distinction between a universal statement that merely happens to be true and one that is true by necessity.

The propositions of geometry are supposed to belong to the second class.

2. NECESSARY TRUTHS

universally true by accident

$\neq$

universally true by necessity

9.3 Definitions

Definitions receive special importance.

Modern thinkers may regard some definitions as conventions about the use of words. The Greeks did not generally understand geometrical definitions in this merely conventional way.

A definition such as a characterization of a circle appeared to state an important truth about circles themselves.

3. DEFINITIONS

Greek view: definition $\neq$ mere abbreviation

definition $\rightarrow$ tells us what the thing is

9.4 Subsumption Under General Rules

Finally, geometrical reasoning repeatedly involves applying a general proposition to a more specific case.

This directs attention toward the relation between general kinds and the particular or subordinate cases falling under them.

4. SUBSUMPTION

specific case falls under general rule

The overall influence of geometry can therefore be summarized compactly.

GEOMETRY ENCOURAGES ATTENTION TO

1. General propositions
2. Necessary truths
3. Definitions
4. Subsumption under general rules

These interests later become prominent in Aristotle.

10 Dialectic and Metaphysical Argument

Greek logic cannot be explained entirely through geometrical demonstration.

A second major source is **dialectic**.

The word is historically important because it was an early technical term for a kind of reasoning connected with philosophical discussion. It is derived from a word associated with discussion or conversation.

In Plato's middle-period dialogues, dialectical procedure often takes the form of examining a hypothesis by tracing its consequences.

If a hypothesis entails an unacceptable consequence, the hypothesis is rejected.

DIALECTIC

Test a hypothesis by its consequences

$$P \rightarrow Q$$

$$\neg Q$$

$$\therefore \neg P$$

Hypothesis $\rightarrow$ unacceptable consequence $\rightarrow$ reject hypothesis

This is the characteristic structure of refutation (*elenchus*).

11 Zeno and Reductio ad Impossibile

The Kneales connect Platonic dialectic both with Socratic refutation and with Zeno of Elea.

Zeno defended Parmenidean monism by drawing absurd consequences from the supposition that plurality exists.

This form of reasoning is commonly described as *reductio ad impossibile*: a hypothesis is provisionally accepted and then rejected because it produces an impossibility.

The method may itself have been encouraged by earlier mathematical practice.

REDUCTIO AD IMPOSSIBILE

Assume P

$\downarrow$

derive impossibility / contradiction

$\downarrow$

$$\therefore \neg P$$

The standard mathematical example discussed by the Kneales is the proof of the incommensurability of the diagonal and side of a square, expressed in modern terminology as the irrationality of $\sqrt{2}$.

12 Worked Example: The Irrationality of $\sqrt{2}$

Suppose, for the sake of argument, that $\sqrt{2}$ is rational.

Then it can be expressed as a ratio

$$\sqrt{2} = \frac{m}{n},$$

where m and n are integers chosen with no common factor.

Squaring gives

$$m^2 = 2n^2.$$

Thus m^2 is even, and therefore m is even.

Because the board is small, the demonstration is best performed in two packets.

REDUCTIO: PACKET 1

Assume

$$\sqrt{2} = \frac{m}{n},$$

where m, n are coprime.

Then

$$m^2 = 2n^2.$$

Therefore

$$m^2 \text{ is even} \quad \Rightarrow \quad m \text{ is even.}$$

So let

$$m = 2k.$$

Copy, then erase before Packet 2.

Substituting $m = 2k$ into the equation gives

$$(2k)^2 = 2n^2.$$

Hence

$$4k^2 = 2n^2$$

and therefore

$$n^2 = 2k^2.$$

So n is also even.

But if both m and n are even, they share a common factor, contrary to the original assumption that they were coprime.

The original hypothesis must therefore be false.

REDUCTIO: PACKET 2

	$m = 2k$
so	
	$m^2 = 2n^2$
becomes	
	$(2k)^2 = 2n^2.$
Thus	
	$4k^2 = 2n^2$
and	
	$n^2 = 2k^2.$
Therefore	
	n is even.
But then	
	m even AND n even,
contradicting the assumption that m, n are coprime.	
$\therefore \sqrt{2}$ is irrational	

The logical structure is more important for us than the mathematical result itself.

ABSTRACT FORM

Assume P
↓
contradiction
↓
$\therefore \neg P$

This example helps connect the mathematical tradition of demonstration with the philosophical tradition of dialectical refutation.

13 Zenonian Reductio and the Socratic Elenchus

The Socratic method resembles Zeno's, but the Kneales point out an important difference.

A strict Zenonian *reductio ad impossibile* produces incompatible consequences or an outright contradiction.

A Socratic refutation need not always reach a formal contradiction. A hypothesis may instead be rejected because it leads to a consequence independently regarded as false or unacceptable.

The *Meno*, for example, contains an argument against the claim that virtue is teachable. If virtue were teachable, good men would be expected successfully to teach it to their sons; the failure of celebrated statesmen to do so is then used against the original thesis.

TWO FORMS OF REFUTATION

Zenonian: $P \rightarrow \text{contradiction} \Rightarrow \neg P$

Socratic: $P \rightarrow Q, \quad Q \text{ false/unacceptable} \Rightarrow \neg P$

Both test a thesis through its consequences.

14 Plato's Changing Use of "Dialectic"

The word "dialectic" does not retain one fixed technical meaning throughout Plato's writings.

In the middle-period dialogues, it is closely associated with the testing of hypotheses through their consequences.

In Plato's later work, the name is transferred to the method of **division and collection**.

Collection is less clearly explained, but division is illustrated as a method for discovering definitions by repeatedly dividing a general notion into alternatives.

PLATONIC "DIALECTIC"

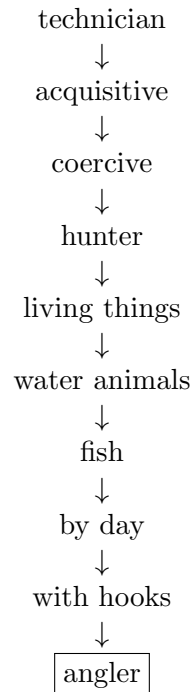
Middle period: hypothesis $\rightarrow$ consequences $\rightarrow$ test

Later period: division + collection $\rightarrow$ search for definition

In the *Sophist*, Plato illustrates division through the attempt to define an angler. A long classificatory tree narrows a general concept by successive choices until the desired definition is reached.

For a small whiteboard, only the selected path needs to be written.

DIVISION: THE ANGLER



The method may appear cumbersome, but it is historically important. The Kneales argue that reflection upon it influenced Aristotle's later development of syllogistic reasoning.

15 From Particular Arguments to Logical Principles

Plato also occasionally states logical principles explicitly in the course of philosophical argument.

The Kneales illustrate the process by imagining a simple exchange.

Suppose someone says:

“Plato was a philosopher, so Plato was vague.”

If challenged, the speaker may supply an unstated general premiss:

“All philosophers are vague.”

The argument now becomes:

All philosophers are vague;

Plato is a philosopher;

$\therefore$ Plato is vague.

MAKE THE PRINCIPLE EXPLICIT

$$\begin{array}{l} \text{All philosophers are vague.} \\ \text{Plato is a philosopher.} \\ \hline \therefore \text{Plato is vague.} \end{array}$$

We can then abstract still further from the subject matter.

ABSTRACT THE FORM

$$\begin{array}{l} \text{Every } M \text{ is } L. \\ S \text{ is } M. \\ \hline \therefore S \text{ is } L. \end{array}$$

Logical form is independent of the particular subject matter.

At this level, the principle no longer depends on whether we are talking about philosophers, vagueness, Plato, or anything else.

Plato sometimes formulates principles of this kind, including a version of the Law of Contradiction.

But he does so piecemeal, as particular arguments require them. He does not assemble the principles into an organized formal system.

Plato: logical principles	=	YES
Plato: systematic formal logic	=	NO

This is one of the major differences between Plato and Aristotle.

16 Eristic, Sophistry, and the Problem of Verbal Form

A third source of early logical reflection comes from ordinary argument, public disputation, and the investigation of fallacies.

An obvious first step toward logic is to notice that different valid arguments seem to share recurring patterns.

But a danger immediately appears: two arguments may look verbally similar while differing radically in validity.

The Kneales use a pair of examples to illustrate the problem.

ARGUMENT 1

This is a pen.
This is blue.
—
∴ This is a blue pen.

Compare:

ARGUMENT 2

That dog is a father.
That dog is his.
—
∴ That dog is his father.

The second argument resembles the first at the surface level of words, but its premisses can be true while its conclusion is false.

The similarity is therefore misleading.

Same-looking words $\neq$ same logical structure

Sophism = fallacious argument resembling a valid one

This is an important step in the discovery that logic cannot simply classify strings of words by superficial grammatical appearance.

17 Public Disputation

Plato and Aristotle provide evidence that formalized public disputation had become a recognizable practice.

Typically, two people participate.

The **answerer** undertakes to maintain a thesis.

The **questioner** attempts to force the answerer to admit consequences that are false, absurd, or otherwise unacceptable.

PUBLIC DISPUTATION

Answerer: maintain a thesis

Questioner: derive an unacceptable consequence

Such disputation could have several purposes.

It might contribute to philosophical investigation.

It might be practiced as intellectual amusement.

It might have practical applications in forensic or political argument.

And the Kneales suggest that at least some apparently trivial sophistical puzzles may have been connected with serious attempts to distinguish valid principles from misleading verbal patterns.

USES OF DISPUTATION

philosophical investigation
amusement
practical argument
logical investigation

18 The Megarian Route Toward Logic

The Kneales suggest that some pre-Aristotelian logical discussion may have been centered in the Megarian school.

Its founder, Euclides of Megara, was associated both with the Socratic tradition and with Eleatic philosophy.

The Megarians became famous for verbal controversy and were sometimes called “Eristics,” but later Megarians such as Diodorus Cronus and Philo displayed genuine logical sophistication.

The Kneales therefore regard it as plausible that apparently eristic disputation supplied material for serious logical reflection.

A SECOND HISTORICAL LINE

debate / sophisms
↓
reflection on argument
↓
Megarian tradition
↓
later Stoic logic

This development is significant because the history of logic does not run through Aristotle alone.

19 Truth, Falsehood, and the Limits of Mere Words

The investigation of sophisms also raises deeper questions about truth and falsehood.

The fragment known as the *Dissoi Logoi* discusses cases in which the same verbal expression can be true when uttered by one person and false when uttered by another.

Consider the verbal form:

“I am an initiate.”

It might be true when spoken by one person and false when spoken by another.

SAME WORDS, DIFFERENT TRUTH VALUE

“I am an initiate.”

spoken by A → TRUE
spoken by B → FALSE

Truth and falsity cannot belong to the verbal pattern alone.

This directs attention toward a question that will become central in the philosophy of logic:

What exactly is the thing that is true or false?

WHAT IS TRUE OR FALSE?

The Kneales suggest that reflection of this sort anticipates the later Stoic distinction between an expression and what is expressed by it.

20 Plato and the Philosophy of Logic

Although Plato does not construct a systematic formal logic, the Kneales regard him as the first great thinker in the **philosophy of logic**.

Three questions organize their discussion of his achievement.

PLATO: THREE QUESTIONS

1. What can properly be TRUE or FALSE?
2. What makes valid inference NECESSARY?
3. What is the nature of DEFINITION?

These questions are not presented by Plato in precisely this later form. The Kneales extract them from discussions in which logical, metaphysical, and epistemological issues remain interwoven.

21 Question One: What Is True or False?

In the *Theaetetus*, Plato's discussion of knowledge leads to the idea of true opinion.

But this raises a further problem: what is an opinion?

Plato sometimes describes thinking as an internal conversation of the soul. Thought is a kind of silent discourse in which one asks, answers, affirms, and denies.

This makes it natural to apply "true" and "false" either to speech itself or to the corresponding act of thought.

In the *Sophist*, Plato again associates thought and discourse and treats complete discourse as involving at least a noun and a verb.

The Kneales therefore detect two possible answers in Plato.

WHAT BEARS TRUTH OR FALSITY?

Plato suggests:

1. sentence / discourse
2. thought / judgement

Later Stoic alternative:

3. proposition

Plato does not produce a fully explicit distinction among these possibilities.

The third answer, associated with the proposition as something distinct from both words and psychological acts, belongs to a later stage of the history.

22 Question Two: What Makes Inference Necessary?

If valid inference consists in tracing necessary connexions, then the previous question immediately leads to another:

What are the things between which necessary connexions hold?

If truth belongs to sentences, one might suppose that necessity is simply a relation between sentences.

If truth belongs to psychological acts, one might suppose that necessity is a relation between thoughts.

The Kneales regard the second suggestion as especially problematic, since people can think without reasoning validly.

Plato's characteristic answer instead appeals to the **Forms**.

PLATO'S ANSWER

Necessary connexions hold between Forms

For the logical point, we need only a limited account of the Theory of Forms.

Forms are not ordinary physical things and are not merely private ideas in a person's mind. They correspond at least partly to what later philosophers call universals: common characters associated with general terms.

Correct thinking follows the connexions among these Forms.

CORRECT THINKING

Forms are connected in reality
↓
thought follows those connexions
↓
true discourse reflects reality

A sentence is true, on this interpretation, when the arrangement expressed in discourse corresponds to the way realities are actually connected.

For Plato, validity is grounded in how realities are connected, not merely in a verbal pattern.

The Kneales note that Plato does not clearly handle every problem generated by this position, including the distinction between singular and general statements.

Nevertheless, the central historical idea is clear: logical necessity is connected with an objective structure rather than merely with arbitrary linguistic habits.

23 Being, Same, and Other

The *Sophist* also contains a discussion of the unusually general notions of **being**, **same**, and **other**. These terms do not operate in the same way as ordinary classifying words such as “red” or “kangaroo.”

They can enter into discourse about almost anything.

Plato’s discussion helped direct attention toward terms that later logical theories would have to treat differently from ordinary predicates.

For the present lecture, however, the most important result is that Plato helps remove two conceptual obstacles inherited from earlier Greek philosophy: difficulties concerning negation and difficulties concerning predication.

24 Negation and Predication

Earlier Greek thinkers had found something puzzling in speaking of what “is not.”

If not-being were simply nothing, then a negative assertion could seem unintelligible.

A related confusion concerned statements of identity and statements of predication.

The sentence

$$A \text{ is } B$$

does not always mean

$$A = B.$$

To say that Socrates is wise, for example, need not mean that Socrates is identical with wisdom.

Plato’s later work helps make room for meaningful negation and non-identical predication.

TWO OLD OBSTACLES

1. NEGATION: A is not B

2. PREDICATION: A is B

but

predication $\neq$ identity

and

A is $B \not\Rightarrow A = B$.

The Kneales emphasize that removing these confusions was important for the later development of logic.

Plato clears conceptual ground for later logical theory.

25 Question Three: What Is a Definition?

Definitions are central throughout Plato's dialogues.

According to Aristotle, Plato inherited part of this concern from Socrates' attempts to define ethical notions. Mathematical practice probably supplied another important influence.

Plato treats definition as both **non-arbitrary** and **informative**.

He recognizes that different languages can use different words for the same thing. The choice of linguistic sign is therefore conventional in an important sense.

But the object of definition is not, for Plato, merely the word.

Definition concerns the thing or Form to which the word refers.

WORD AND DEFINITION

word $\rightarrow$ conventional sign

definition $\rightarrow$ thing / Form

Thus the question

"What is justice?"

is not merely the question

“How do Greek speakers happen to use the word *justice*?”

It asks instead for the common nature that makes the relevant things instances of justice.

This is the historical source of what is called a **realistic theory of definition** or a **real definition**.

REAL DEFINITION
Not merely:

“How do we use this word?”

but:

“What is the thing?”

Examples:

What is justice?

What is colour?

For Plato, definition is non-arbitrary and informative.

This conception will strongly influence Aristotle even though Aristotle rejects Plato’s separate Forms.

26 How the Three Platonic Questions Fit Together

The Kneales’ three questions are closely connected.

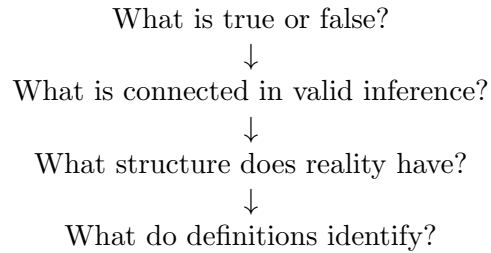
If logic concerns valid inference, validity involves necessary connexion.

To understand necessary connexion, we must ask what sorts of things are connected.

That question in turn requires us to ask what sort of thing can be true or false.

And Plato’s theory of definition supplies part of his answer to how general thought connects with the structure of reality.

PLATO'S LOGICAL PROBLEMS



In Plato, therefore, logic has not yet separated cleanly from metaphysics and epistemology.

That separation will itself be part of the later history.

27 Was There Logic Before Aristotle?

The evidence surveyed by the Kneales supports a qualified answer.

There was certainly substantial logical reflection before Aristotle.

Greek geometers had reflected upon demonstration, axioms, necessity, and consequence.

Zeno and Socrates had developed methods of refuting hypotheses through their consequences.

Sophistical and eristic argument directed attention toward the difference between genuinely valid structure and misleading verbal appearance.

The Megarian tradition seems to have continued serious work on logical puzzles.

Plato formulated some logical principles and investigated major philosophical problems concerning truth, necessary connexion, negation, predication, and definition.

BEFORE ARISTOTLE

Geometry	→	demonstration
Zeno / Socrates	→	refutation
Sophists / Megarians	→	fallacy and argument
Plato	→	truth, necessity, definition

But something crucial is still missing.

These investigations do not yet constitute a systematic formal theory of inference.

Plato states logical principles as he requires them, but does not organize them into a connected logical system.

The Kneales therefore preserve the magnitude of Aristotle's achievement while rejecting the idea

that logical reflection began from nothing with Aristotle.

Much logical reflection existed before Aristotle,
but no systematic formal logic yet.

ARISTOTLE'S DECISIVE ACHIEVEMENT: SYSTEMATIZATION

28 Lecture Summary

The history traced in this chapter is not the story of a single invention at a single moment.

Logical theory emerges gradually when Greeks begin reflecting upon practices of proof and argument that already existed.

Geometry produces the ideal of necessary demonstration from explicit starting points.

Metaphysical and Socratic dialectic develop the examination of hypotheses through their consequences.

Sophistical and eristic disputes expose the danger of confusing verbal similarity with valid logical form.

Plato then raises fundamental questions about truth and falsity, necessary connexion, predication, negation, and definition.

These developments create much of the intellectual material from which Aristotle will construct the first systematic theory of formal logic.

SUMMARY

Reasoning
↓
reflection on validity
↓
logic

Geometry → demonstration / consequence
Dialectic → hypothesis / refutation
Sophistry → validity vs. verbal appearance
Plato → truth / necessity / definition

Pre-Aristotelian Greeks supplied the problems and methods; Aristotle
supplied systematic formal theory.